



BITS Pilani
Pilani | Dubai | Goa | Hyderabad

Work Integrated Learning Programmes Division

M.Tech. in AIML

Natural Language Processing

S2-25_AIMLCZG530

Assignment 2

Problem Statement-11

General Instructions

- 1. Implementation on the BITS OSHA Virtual Lab is compulsory, and carries 1 mark.**
- 2. Submit the code in .pdf/html format along with output. Please follow the naming convention as <Group no>.pdf. Example for group 1 should be named as - Group1.pdf.**
- 3. Include group details like team members' names, BITS ID, and contribution percentages.**
- 4. All members of the group will work on the same problem statement. Each group should upload in Taxila in respective locations under ASSIGNMENT Tab. Assignments submitted via means other than through Taxila will not be graded.**
- 5. One student per group is requested to submit your assignment.**
- 6. Any queries related to this problem statement should be addressed to Vasugi I (vasugii@wilp.bits-pilani.ac.in), Course LE.**
- 7. If the dataset link is not working, download a similar dataset from an online resource and note this in your submission.**

Customer Review Text Summarization

Problem Statement

E-commerce platforms receive thousands of customer reviews for products, making it challenging for buyers to understand overall customer opinions. Design and implement an **end-to-end Customer Review Summarization Application** that generates concise summaries of multiple customer reviews, highlighting the most frequently discussed positive and negative aspects. The application should use an Encoder–Decoder model to produce meaningful summaries through a web-based interface.

Tasks to perform

1. **Problem Analysis (2 Marks)**
 1. Identify the application domain.
 2. Define the problem statement and functional requirements.
 3. Specify the expected input and output of the application.
2. **Data Collection and Preprocessing (2 Marks)**
 1. Select a suitable text summarization dataset or collect sample documents.
 2. Perform text preprocessing, including tokenization, cleaning, and normalization.
 3. Prepare the dataset for model training and testing.
3. **Model Development (3 Marks)**
 1. Build an Encoder–Decoder (Seq2Seq/LSTM or Transformer) model.
 2. Train the model to generate abstractive summaries.
 3. Generate summaries for the given input documents.
4. **Application Development (3 Marks)**
 1. Develop a web application using Streamlit or Flask.
 2. Provide options for text input or document upload.
 3. Display the generated summary through the application interface.
5. **Evaluation and Demonstration (2 Marks)**
 1. Assess the generated summaries using ROUGE or other suitable evaluation metrics.
 2. Demonstrate the application using sample inputs and outputs.
 3. Present observations based on the generated summaries.
6. **Project Report (1 Mark)**
 1. Include implementation details, screenshots, observations, conclusion, and references in a well-structured report. (include as a pdf)